

Agenda

- ① About C language
- ② printf()
- ③ Write your first program.
- ④ Escape Sequences
- ⑤ Format Specifiers
- ⑥ scanf()

About C language

- ① C is a block structured programming language
- ② A C program can have any number of blocks
- ③ Usually outermost blocks are functions
- ④ function has some name for identification
- ⑤ Even the smallest C program has one function
- ⑥ All function names must be different
- ⑦ If there is only one function its name must be `main()`
- ⑧ If there are more than one function in the program then also one function name must be `main`.

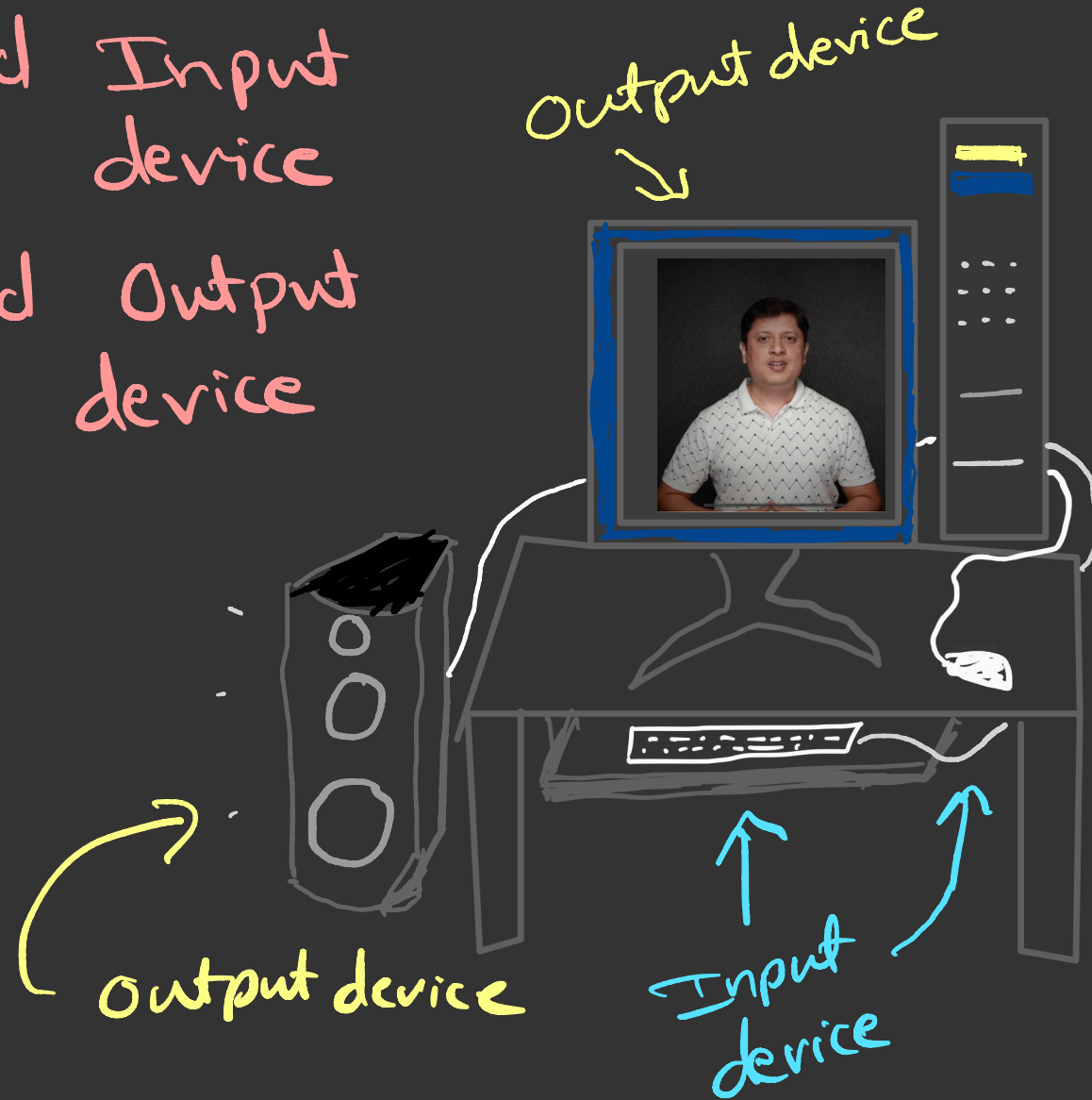
About C language

- ⑨ You can write declaration statements inside a block (function) or outside all functions.
- ⑩ Local variables vs Global variables
- ⑪ Action statements can reside only inside the block (function)
- ⑫ C is a case sensitive language.
`int a;` ← lower case
`A=5;` error
 ← upper case

Standard Input & Output devices

Keyboard → standard Input device

Monitor → standard Output device



Output Instruction

printf() → predefined function

printf() is used to print some text on the monitor.

printf("Hello");

printf()

① to print text

② to print value of a variable or an expression.

```
printf()  
{  
    ==  
    ==  
    ==  
}
```

Write a program to print **welcome** on the screen.

```
#include <stdio.h>
```

preprocessor

```
int main()
```

Compiler

```
{
```

```
    printf("welcome");
```

Linker

```
    return 0;
```

```
}
```

Write a program to print **welcome students** on the screen. Print **welcome** on the first line and **Students** on the second line.

```
#include <stdio.h>
int main()
{
    printf("Welcome\nStudents");
    return 0;
}
```

Escape Sequences

<code>\n</code>	new line
<code>\t</code>	tab space
<code>\b</code>	back space
<code>\r</code>	carriage return
<code>\f</code>	form feed
<code>\\</code>	print \
<code>\"</code>	print "
<code>'</code>	print '

Write a program to print value of a variable.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

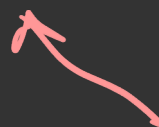
```
    int a = 5;
```

```
    printf ("%d", a);
```

```
    return 0;
```

```
}
```

Format String



Format Specifiers

%d int

%c char

%f float

%lf double

Write printf statement to print following on the screen.

Consider $\rightarrow$ `int a=5, b=6;`

① 5 6 `printf("%.d %.d", a, b);`

② `a=5 and b=6` `printf("a=%.d and b=%.d", a, b);`

③ `a=6 & b=5` `printf("a=%.d & b=%.d", b, a);`

④ value of a is 5 and value of b is 6 $\rightarrow$ expression $\rightarrow$

⑤ sum of 5 and 6 is 11 $\rightarrow$ `printf("Sum of %.d and %.d is %.d", a, b, a+b);`

⑥ `5+6=11`

`printf("Value of a is %.d and value of b is %.d", a, b);`

$\rightarrow$ `printf("%.d + %.d = %.d", a, b, a+b);`

scanf()

scanf() → predefined function

scanf("%d", &a);

↖ address of

Write a program to calculate sum of two integers.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b, c;
```

```
    printf("Enter two numbers");
```

```
    scanf("%d %d", &a, &b);
```

```
    c = a + b;
```

```
    printf("Sum of %d and %d is %d", a, b, c );
```

```
    return 0;
```

```
}
```